Circle configurations in classical plane geometry, formalized in Lean 4: Clifford's chain, Miquel's five circles, Lester, Sawayama–Thébault and Kosnita

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Abstract

We give machine-checked proofs, in Lean 4 with Mathlib, of a family of classical theorems about configurations of circles in the Euclidean plane. The centrepiece is the beginning of Clifford's chain: for four circles through a common point in general position, the four circles through triples of their second intersection points have a common point (the case $n = 4$); for five such circles the five points so obtained lie on one circle *or one line* (the case $n = 5$); and, under a stated further non-degeneracy condition, for six circles the six circles so obtained have a common point (the case $n = 6$). We also prove Miquel's five-circles theorem for the pentagram (again with the conclusion "on a circle or on a line"), Lester's theorem that the circumcentre, the nine-point centre and the two Fermat points of a scalene triangle are concyclic, the Sawayama–Thébault theorem on the centres of the two Thébault circles and the incentre, and Kosnita's concurrence theorem; and, as two short companions in the same style, Blundon's inequalities and the Steiner–Lehmus theorem. Every degenerate configuration that a statement excludes is named, and for several exclusions we exhibit configurations showing that the hypothesis cannot simply be dropped. All proofs use only Lean's three standard axioms. In the sources and search scope listed in Section 10 we found no previous complete formal proof of any of these theorems, with one partial exception: the algebraic core of Steiner–Lehmus (over side lengths, with the bisector length given by its formula) has a Lean proof, and we found no formal proof of the geometric theorem. The formal proofs we found closest to the circle theorems are of the three-circle Miquel theorem, which is the base case of Clifford's chain. We also show, with an explicit six-circle configuration, that the natural finite-plane formalization of the chain is false for every $n \geq 6$: its Clifford point lies at infinity.

1 Introduction

Many of the most attractive theorems of nineteenth-century plane geometry are *incidence theorems for circles*: some construction is carried out with circles, and at the end several points turn out to lie on one circle, or several circles turn out to pass through one point. The simplest is Miquel's

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pivot theorem [1]: if a point is chosen on each side of a triangle, the three circles through a vertex and the two chosen points on the adjacent sides pass through a common point. It is formalized in Isabelle/HOL [16], both as an incidence statement and with existence of the common point; Rocq's HighSchoolGeometry [17] has the incidence form of the related four-line (complete quadrilateral) Miquel theorem.

Clifford [2] discovered that Miquel's theorem is the first step of an infinite chain. Take n circles in general position through a common point M. Any two meet again in a point; any three determine a circle through their three such points; for four circles, the four circles so obtained are concurrent; for five, the five points so obtained are concyclic; for six, the six circles so obtained are concurrent; and so on, alternating between points and circles. Miquel's five-circles theorem, the Sawayama–Thébault theorem and Kosnita's theorem are further classical circle configurations, treated in the standard texts [3, 4] and, for Thébault's problem, by computer algebra [15]; Lester's theorem is more recent [5, 6]. In the sources we searched (Section 10) we found no complete formal proof of any of them in a proof assistant.

1.1 Contributions

All results below are proved in Lean 4 [24] with Mathlib [25].

1. **Clifford's chain,** $n = 4, 5, 6$ (Theorems 1–3), in an explicit complex-coordinate model, together with a second, independent development (Section 2.3) which defines an existential predicate for the chain at every n, proves the cases $n = 4, 5$ in that general form ($n = 5$ with no hypothesis beyond general position), and proves that it fails for every $n \geq 6$: an explicit six-circle configuration in general position has its Clifford point at infinity, where the finite plane has no room for it.

2. **Miquel's five-circles theorem** (Theorem 6).

3. **Lester's theorem** (Theorem 7).

4. **The Sawayama–Thébault theorem** (Theorem 8).

5. **Kosnita's theorem** (Theorem 9).

6. **Blundon's inequalities** and **the Steiner–Lehmus theorem** (Theorems 10 and 11), which are not circle configurations but were proved by the same method in the same project.

Method. The proofs follow the method of the earlier papers in this series [26, 27]: choose coordinates (usually complex), reduce the geometric conclusion to the vanishing of a polynomial in the coordinates, and prove that vanishing in Lean by an explicit algebraic certificate (`linear_combination`, `field_simp`, `nlinarith`); the exception is the five-circles theorem, whose core lemma is an argument with doubled directed angles (Section 3). The certificates come with separate lemmas tying the coordinate objects to Mathlib's notions (circumcentre, incentre, nine-point circle, `Cospherical`, `Concyclic`, `Collinear`). They are large (the $n = 6$ Clifford case is the largest) but entirely mechanical; what needs care is the *statements*, and in particular their non-degeneracy hypotheses.

What is not claimed. We do not prove Clifford's chain for general n, and we show that the natural finite-plane statement of it is false for $n \geq 6$ (Sections 2.3 and 2.4). Every theorem is stated for configurations satisfying explicit non-degeneracy hypotheses; Section 8 lists them and

says which cannot simply be dropped. Theorems are stated in the Euclidean plane, not in the inversive (Möbius) plane, so "circle" means a Euclidean circle, and lines appear only as stated alternatives.

1.2 Availability

The proofs are in eight Lean 4 developments on Lean v4.33.1 with Mathlib pinned at revision `0df444a3` (all eight lockfiles agree). File and line counts below are of the library `.lean` files, including the root import file, counting every physical line (comments and blank lines included) and excluding scratch files; in total 24 files and 5,049 lines.

development	content	files	lines
`clifford`	Clifford $n = 4, 5, 6$	3	1,271
`cliffordchain`	the chain predicate: $n = 4, 5$; fails for $n \geq 6$	7	1,679
`fivecircles`	Miquel's five circles	2	428
`lester`	Lester's theorem	4	506
`thebault`	Sawayama–Thébault	2	559
`kosnita`	Kosnita's theorem	2	249
`blundon`, `steinerlehmus`	Blundon; Steiner–Lehmus	2+2	228+129

They are published on GitHub; the paper is checked against these commits, and every source reference (`File`:*line*) links to that line at the commit:

development	repository	commit
`clifford`	`https://github.com/jpbald93/clifford`	`209abaa5e638`
`cliffordchain`	`https://github.com/jpbald93/cliffordchain`	`ba083b2158e1`
`fivecircles`	`https://github.com/jpbald93/fivecircles`	`f1fd9b7b2bd9`
`lester`	`https://github.com/jpbald93/lester`	`9f94754987cb`
`thebault`	`https://github.com/jpbald93/thebault`	`c37f92e53c9e`
`kosnita`	`https://github.com/jpbald93/kosnita`	`702f0263a701`
`blundon`	`https://github.com/jpbald93/blundon`	`118b4c9f3afd`
`steinerlehmus`	`https://github.com/jpbald93/steinerlehmus`	`a3a3d26bee6c`

Each development has a gate script that rejects `sorry`, `admit`, `native_decide`, `axiom` and meta-programming commands in the sources, builds the library, and checks with `#print axioms` that every listed declaration depends only on `propext`, `Classical.choice` and `Quot.sound`; definitions and lemmas they depend on are covered transitively. Every declaration cited with a qualified name in this paper, such as `Clifford.clifford_one`, is in its development's list; other names (in plain typewriter type) are auxiliary definitions or lemmas covered in that way. The gate runs in its strict default mode (a permissive `SCAFFOLD=1` mode exists for development and was not used).

2 Clifford's chain: $n = 4, 5, 6$

2.1 The model

We identify the plane with $\mathbb{C}$ and put the common point M at the origin. A circle through 0 is determined by its centre $c \neq 0$: it is the circle $|z - c| = |c|$, i.e. $z\bar{z} = \bar{c}z + c\bar{z}$ (`Clifford.circ_throughOrigin`). For two such circles with distinct centres $c_i \neq c_j$, define

$$P_{ij} \;=\; \frac{\bar{c}_i\, c_j - c_i\, \bar{c}_j}{\bar{c}_i - \bar{c}_j}$$

(`secondInter`, `Basic`:194); for example $c_1 = 1$, $c_2 = i$ give $P_{12} = 1 + i$. Lean proves that P_{ij} lies on both circles (`Clifford.secondInter_mem_left`, `Clifford.secondInter_mem_right`). It is the second intersection point when it differs from 0; it equals 0 exactly when the circles are tangent at 0 (e.g. $c_1 = 1$, $c_2 = 2$), which the theorems below exclude by assuming $P_{ij} \neq 0$. For three circles i, j, k, Γ_{ijk} (`gam`, `Basic`:203) is the circumcircle of P_{ij}, P_{ik}, P_{jk}, with centre given by an explicit circumcentre formula (`ccenter`, `Basic`:197); when the three points are not collinear Lean proves that they lie on it (`Clifford.mem_circumcircle_left` and its two companions). Mathlib's `EuclideanGeometry.Sphere` is used for circles throughout.

2.2 The theorems

Theorem 1 (Clifford, $n = 4$; `Clifford.clifford_one`)**.** *Let four circles pass through a common point M, with centres $c_1, \ldots, c_4$. Suppose every second intersection point P_{ij} $(i < j)$ differs from M, and for each triple $\{i, j, k\}$ the three points P_{ij}, P_{ik}, P_{jk} are not collinear. Then the four circles $\Gamma_{123}, \Gamma_{124}, \Gamma_{134}, \Gamma_{234}$ have a common point.*

The theorem (`Basic`:381) asserts existence only. Its witness is an explicit point, a ratio of two determinant-like polynomials in the centres and their conjugates, $\mathrm{cp}(c_1, c_2, c_3, c_4)$ (`cliffordPoint`, `Basic`:309); Lean proves that it lies on each Γ_{ijk} (`Clifford.cliffordPoint_mem_gam`, `Basic`:323, and `Clifford.cliffordPoint_mem_gams` for all four at once). We do not prove that the common point is unique, and the theorems below are about these explicit points.

Theorem 2 (Clifford, $n = 5$; `Clifford.clifford_two`, `Clifford.clifford_two_explicit`)**.** *Let five circles pass through M, with every $P_{ij} \neq M$ and no triple P_{ij}, P_{ik}, P_{jk} collinear. For each k, let $P_k = \mathrm{cp}$ of the four circles other than circle k. Then each P_k lies on the four circles Γ of those four, and the five points $P_1, \ldots, P_5$ lie on one circle or on one line.*

In Lean the statement is split in two. `Clifford.clifford_two` (`Third`:543) asserts that there *exist* five points with these incidences that lie on a circle or a line; `Clifford.clifford_two_explicit` (`Third`:457) proves the circle-or-line conclusion for the five explicit points cp, and `Clifford.cliffordPoint_mem_gams` gives their incidences. The disjunction is not a weakening: in the independent agent's numerical check (reported figures, script not kept), configurations satisfying every hypothesis were found in which the five points are pairwise distinct (minimum spacing 0.042) and collinear to 5×10^{-60}. The second alternative corresponds to the usual inversive formulation, in which a line is a circle through ∞; our statement is the finite Euclidean circle-or-line form of it, under the stated hypotheses.

Theorem 3 (Clifford, $n = 6$; `Clifford.clifford_three`)**.** *Let six circles pass through M, with every $P_{ij} \neq M$ and no triple P_{ij}, P_{ik}, P_{jk} collinear. For each five-element subset S of the six, let $P_T^S = \mathrm{cp}(c_T)$ for the five four-element subsets $T \subset S$, and let C_S be the circle through three designated ones of them (for $S = \{1, \ldots, 5\}$: those of $\{1,2,3,4\}$, $\{1,2,3,5\}$, $\{1,2,4,5\}$; the six choices are listed in the Lean statement). Suppose those three points are not collinear, for each S. Then all five points P_T^S lie on C_S, for each S, and the six circles C_S have a common point.*

The six extra hypotheses (`m1`–`m6` in `Third`:780) say that the three designated points of each S are not collinear, so that C_S is a genuine circle; they exclude exactly the case that Theorem 2 allows in its second alternative (and the case of coincident points). So Theorem 3 is narrower than the generalized-circle statement: it is Clifford's third theorem for configurations whose five-point circles are honest circles. The proof reduces all six memberships to one polynomial identity (`Clifford.master`, `Third`:187), the largest certificate in this paper.

2.3 The chain for general n

A second, independently written development, `cliffordchain`, defines an existential incidence predicate for the whole chain. Its *generalized circles* (`IsGCircle`, `Clean`:156) are lines and Mathlib spheres `Metric.sphere` $o\,r$ for any real r; so a radius-0 "circle" (a single point) and, formally, the empty set ($r < 0$) are included. The proved case $n = 5$ uses the radius-0 case when all five points coincide. For centres $c : \{0, \ldots, n-1\} \to \mathbb{C}$, the predicate $\mathrm{Chain}(n, c, X)$ (`Clean`:165) says that X is a Clifford object of level n: for $n = 0, 1, 2$ the point M, the circle itself, and the point P_{01}; for odd $n \geq 3$, a generalized circle containing, for each i, *some* level-$(n-1)$ point of the circles other than i; for even $n \geq 4$, a single point lying, for each i, on *some* level-$(n-1)$ object of the circles other than i. The predicate asserts existence at each level; it does not assert that the objects are unique or that the choices at different levels cohere. The development's intended general statement is CliffordChain(n): in general position (pairwise transversal, every triple non-degenerate) a level-n object exists (`Clean`:184). The general-position conditions are stated as polynomial inequalities, and a bridge file ties them to geometry (`Bridge`:58 onwards): `Transversal` $c\,d$ holds if and only if M, c, d are not collinear (`Clifford.transversal_iff`); the first condition of `GenTriple` holds if and only if the three centres are not collinear (`Clifford.triP_ne_zero_iff`); and, for pairwise transversal circles with non-collinear centres, the second holds if and only if the three second intersection points are not collinear (`Clifford.cycP_ne_zero_iff`). The last needs the non-collinear centres: the underlying identity is $X \prod |c_i - c_j|^2 = \mathrm{triP}^2 \cdot \mathrm{cycP}$, where X is the collinearity determinant of the three points. Together, `Clifford.genTriple_iff` says that `GenTriple` means exactly "neither the centres nor the three second intersection points are collinear"; and `Clifford.onCircleM_iff_mem_sphere` identifies the circle model with Mathlib's `Metric.sphere`.

Theorem 4 (`Clifford.cliffordChain_four`, `Clifford.cliffordChain_five`)**.** CliffordChain(4) *and* CliffordChain(5) *hold.*

The $n = 5$ case rests on `Clifford.clifford_two_clean` (`Clean`:123): in general position the five points are concyclic or collinear, with no further hypothesis. (The same development also contains a version of the $n = 5$ theorem with an extra coefficient condition, `CliffordNondeg`, which excludes some genuine configurations; we do not cite that version as Clifford's theorem.) For $n \geq 6$ the predicate is in fact false; see Section 2.4.

2.4 The finite-plane chain fails for $n \geq 6$

The predicate Chain places every Clifford object in the finite plane $\mathbb{C}$. Classically, Clifford's theorem lives in the inversive plane: after inversion at M the n circles become n lines in general position, whose Clifford objects are honest points and circles, and the inverse image of a Clifford point that happens to be M itself is the point at infinity. Nothing in "pairwise transversal, every triple non-degenerate" prevents this, and the finite-plane predicate has no point at infinity to use. We prove that this already happens for six circles, and therefore for every $n \geq 6$.

Theorem 5 (`Clifford.Counter6.not_cliffordChain_six`, `Clifford.not_cliffordChain_of_six_le`, `Clifford.cliffordChain_iff`)**.** *For the six centres*

$$c = (3 - 6i,\ 6 - 3i,\ 1 + 2i,\ -2 - i,\ -4i,\ 4)$$

the circles through $M = 0$ are in general position in the sense of CliffordChain*, but no level-*6 *object exists. Consequently* CliffordChain(n) *fails for every $n \geq 6$, and* CliffordChain(n) *holds if and only if $n \leq 5$.*

Number the centres $c_1, \ldots, c_6$ in the order displayed. For this configuration every level-4 object is forced (by the three circles through it) to be the classical Clifford point, and for each of the six five-circle subfamilies the five such points are distinct and *collinear*, so the level-5 objects are forced to be lines: for instance $\operatorname{Re} z = 4$ when circle 5 is omitted and $\operatorname{Im} z = -4$ when circle 6 is omitted. These two lines meet at $4 - 4i$, which is not on the line for omitted circle 1, so no point lies on all three, let alone all six, level-5 objects (`Counter6`:182). The kernel checks what the proof needs: general position, the nine level-4 points that determine these three lines, and their non-concurrence, all in exact arithmetic over $\mathbb{Q}(i)$. The full picture (all fifteen level-4 points and all six lines) was checked outside Lean in exact rational arithmetic, by the agent and independently by us. Replacing $c_6 = 4$ by $4 + \varepsilon$ (real $\varepsilon \neq 0$, small enough that the configuration stays in general position) gives, in exact arithmetic, the classical level-6 point $4 - 4i + 50/\varepsilon$, which goes to infinity as $\varepsilon \to 0$. The step to $n \geq 7$ uses that a level-$(n+1)$ object requires a level-n object of the subfamily omitting the last circle, together with the fact that any family in general position can be extended by one more circle in general position (the new centre is taken on a parabola, which meets each excluded locus in finitely many points); see `Classify`:57 and `Classify`:167.

Together, Theorems 4 and 5 settle the predicate completely: it is a correct formalization of the cases $n = 4, 5$ but not of Clifford's theorem for $n \geq 6$. A faithful general statement needs either the Riemann sphere (even-level objects in $\mathbb{C} \cup \{\infty\}$, lines counted as circles through ∞) or the equivalent formulation for n lines. In the same exploration (60 digits, random centres) the classical chain exists for $n \leq 8$ and agrees with a closed form after inversion, but we have not formalized either corrected statement. The six-circle theorem of Section 2 (Theorem 3) is unaffected: its hypotheses m_1–m_6 require the designated triples to be non-collinear, and in this configuration every five-point set is collinear (an argument, not a machine-checked statement).

3 Miquel's five circles

Take a pentagon $A_0, \ldots, A_4$, extend its sides to form a pentagram with outer points $T_0, \ldots, T_4$, and draw the five circles through A_{i-1}, A_i and the triangle point T_i. Consecutive circles meet again at points M_i. Miquel's theorem [1] says that the five points M_i are concyclic.

Theorem 6 (`Fivecircles.fivecircles`)**.** *Let $A, T, M : \{0, \ldots, 4\} \to \mathbb{C}$ be injective, with no point of one family equal to a point of another (so the fifteen labelled points are pairwise distinct). Suppose that for each i (indices mod 5) the points $A_{i-1}, A_i, T_{i+1}, T_{i-1}$ are collinear and the points $A_{i-1}, A_i, T_i, M_{i-1}, M_i$ are concyclic. Then $M_0, \ldots, M_4$ lie on one circle or on one line.*

The collinearity hypotheses say that $T_{i\pm1}$ lie on the line $A_{i-1}A_i$, which is the pentagram incidence; the concyclicity hypotheses say that M_{i-1} and M_i are the second intersections of the circle of triangle i with its neighbours. Distinctness of the fifteen points excludes, in particular, tangent circles (a second intersection coinciding with A_i). Again the alternative is genuine: the independent agent's numerical check (reported figures, script not kept) found a non-degenerate pentagram (minimum point separation 0.52) whose five points M_i are collinear to 10^{-39}. Unlike the other developments, the core of this proof is not polynomial algebra. Concyclicity of four points is reduced to the vanishing of the imaginary part of a cross-ratio, and that to an equality of doubled directed angles (`Real.Angle`); the core lemma (`Fivecircles.core`, `Basic`:167) is then an angle chase, written as a sequence of additive certificates, and is reused for the five cyclic relabellings. The theorem is `Basic`:383.

4 Lester's circle

Theorem 7 (Lester; `Lester.lester`)**.** *Let ABC be a scalene triangle in the plane. Let F_1 be its first Fermat point, the common point of the three lines joining each vertex to the apex of the equilateral triangle erected outwards on the opposite side, and F_2 its second Fermat point (the same with the equilateral triangles erected inwards). Then the circumcentre, the nine-point centre, F_1 and F_2 are concyclic.*

The theorem is due to Lester [5]; see [6] for the circles of Lester, Evans and Parry. The hypothesis is *scalene*, not merely "not equilateral". For an isosceles triangle all four points lie on the axis of symmetry, and for a typical one they are four distinct points: for $A = (0, h)$, $B = (-1, 0)$, $C = (1, 0)$ with $h \in \{0.3, 0.5, 1, 2.5\}$ we computed (60 digits, archived script) four distinct collinear points (minimum separation 0.048 to 0.33), which are not concyclic. So the scalene hypothesis cannot simply be weakened to "not equilateral". It does more than is needed, though: in some isosceles triangles points coincide and the conclusion holds trivially, since `Concyclic` is a property of a *set*. For the isosceles triangle with apex angle 120°, $A = (0, 1/\sqrt{3})$, the nine-point centre and the first Fermat point both equal A, and the second Fermat point equals the circumcentre, so the set of four points has only two elements. The circumcentre is Mathlib's; the nine-point centre is defined as the midpoint of the circumcentre and the orthocentre and proved equal to the centre of Mathlib's nine-point circle (`Lester.bridge_ninePointCenter`); the equilateral apexes are proved equilateral (`Lester.bridge_apexOut`, `Lester.bridge_apexIn`); and $e^{i\pi/3}$ is identified with the Cartesian constant used (`Lester.bridge_ÏĽ`). The theorem is `Basic`:56. It assumes the Fermat points and does not assert their existence.

5 The Sawayama–Thébault theorem

Thébault proposed in 1938 [8] the following problem, which had been proved already by Sawayama in 1905 [7]; see [9] for its history.

Theorem 8 (Sawayama–Thébault; `Thebault3.sawayama_thebault`)**.** *Let ABC be a triangle and D a point strictly between B and C. Let P be the centre of a circle tangent to the lines AD and BC, internally tangent to the circumcircle of ABC, and lying in the angle $\angle ADB$; let Q be the centre of such a circle lying in the angle $\angle ADC$. Then P, Q and the incentre of ABC are collinear.*

The definition `IsThebaultCircle` (`Basic`:32) says that there is $r > 0$ with P at distance r from both lines, at distance $R - r$ from the circumcentre, on the same side of AD as B (resp. C) and on the same side of BC as A; the two side conditions together are exactly "in the open angle". In the independent numerical check, each angle contained exactly one such circle, so the definition picks out the intended circle. The incentre is Mathlib's, and `Thebault3.tri_incenter_spec` proves it equidistant from the three side lines and on the inner side of each. "Exactly one such circle" is the agent's reported observation on 1989 sampled triangles (script not kept), not a proved uniqueness statement. The theorem assumes P and Q and does not assert their existence. It is `Basic`:514.

6 Kosnita's theorem

Theorem 9 (Kosnita; `Kosnita.kosnita`)**.** *Let ABC be a triangle with circumcentre O, and let O_a, O_b, O_c be the circumcentres of the triangles OBC, OCA, OAB. Suppose that no two of A, B, C are diametrically opposite (so that OBC, OCA, OAB are triangles) and that $AB^2 + BC^2 + CA^2 \neq 5R^2$, where R is the circumradius. Then the lines AO_a, BO_b, CO_c are concurrent.*

The circumcentres are specified by equidistance (`hOa1`, `hOa2`, ...), which in the plane is equivalent. The first exclusion (`hOBC`, `hOCA`, `hOAB`: O, B, C not collinear, etc.) excludes right triangles, for which O is the midpoint of a side and O_a does not exist. In the Lean statement it is in fact redundant: if O, B, C were collinear, then O would be the midpoint of BC, and no point is equidistant from three distinct collinear points, so the equidistance hypotheses for O_a could not hold. We keep it because the proof uses it. The second, `hS`, cannot simply be dropped. The concurrence point (classically the Kosnita point, X_{54} in Kimberling's encyclopedia [11]; our theorem does not identify it) can fail to exist on the locus $AB^2 + BC^2 + CA^2 = 5R^2$. In the independent agent's numerical check, three sampled triangles on that locus gave three pairwise parallel, distinct lines (determinants of the line pairs about 10^{-61}, separations about 0.18; reported figures, script not kept), so for them no common point exists and the conclusion fails. Not every triangle on the locus behaves this way. For the isosceles triangle $A = (1, 0)$, $B, C = (\frac{1}{2}, \pm\frac{\sqrt{3}}{2})$ on the unit circle (sides $1, \sqrt{3}, 1$, angle 120° at A), $O_a = A$, so "the line AO_a" is undefined, while BO_b and CO_c are the same line; here the Lean conclusion holds trivially (any point of that line will do). So we do not claim a uniform description of the excluded locus; only that the affine conclusion fails on part of it. Relative to a projective statement "concurrent, possibly at infinity" the exclusion is a weakening. The theorem is `Basic`:180; see [10] for the Kosnita point.

7 Two companions: Blundon and Steiner–Lehmus

Theorem 10 (Blundon; `Blundon.blundon`)**.** *For a triangle in a real inner product space with circumradius R, inradius r and semiperimeter s,*

$$2R^2 + 10Rr - r^2 - 2(R - 2r)\sqrt{R^2 - 2Rr} \;\le\; s^2 \;\le\; 2R^2 + 10Rr - r^2 + 2(R - 2r)\sqrt{R^2 - 2Rr}.$$

Blundon proved that these bounds are best possible [12]. R and r are Mathlib's `Affine.Simplex.circumradius` and `Affine.Simplex.inradius`. The proof is a single polynomial certificate (`Blundon.core_identity`): with x, y, z the tangent lengths, $s^3(4r^2E + \Delta^2) = 0$ where $E = s^4 - 2(2R^2 + 10Rr - r^2)s^2 + r(4R + r)^3$ and Δ is the product of the differences of the sides; hence $E \le 0$, which is the stated pair of inequalities. Bridges tie R and r to the side lengths (`Blundon.circumradius_sq_mul`, `Blundon.height_sq_mul`). The theorem is `Basic`:147.

Theorem 11 (Steiner–Lehmus; `SteinerLehmus.steiner_lehmus`)**.** *In a non-degenerate triangle ABC, let E on segment AC and F on segment AB be the feet of the internal bisectors from B and C. If $BE = CF$ then $AB = AC$.*

The bisector-length formula $BE^2(BA + BC)^2 = BA \cdot BC\,((BA + BC)^2 - AC^2)$ is proved from Mathlib's angles (`SteinerLehmus.bisector_foot_sq`); the two cleared formulas then force $b = c$ through the factorisation $(b - c) \cdot a(a + b + c)(a^3 + a^2b + a^2c + 3abc + b^2c + bc^2) = 0$ (`SteinerLehmus.algebraic_core`). The theorem holds in any real inner product space, and is `Basic`:112. (The proof is direct; the theorem's notoriety for lacking a direct proof [13, 14] concerns synthetic proofs.) The algebraic step alone, $b = c$ from equality of the two bisector-length *formulas* over positive side lengths, has an earlier sorry-free Lean proof [19]; what is new here is the geometric statement, with the bisector feet given as points and the bisector condition as an equality of Mathlib angles, and the bridge `SteinerLehmus.bisector_foot_sq` from it to the formula.

8 Non-degeneracy: what each statement excludes

theorem	excluded configurations	status
Clifford $n = 4, 5, 6$	tangent circles or equal centres ($P_{ij} = M$); a Γ_{ijk} that is a line (three collinear P's)	standard general position
Clifford $n = 5$	— (conclusion: circle *or* line)	line case genuine
Clifford $n = 6$	three designated points of some C_S collinear or coincident	narrower than the inversive statement
Five circles	any coincidence among the fifteen labelled points, incl. tangent circles	line case genuine
Lester	isosceles triangles	cannot be weakened to "not equilateral"
Sawayama–Thébault	D at B or C; degenerate triangle	classical setting
Kosnita	right triangles (redundant given the other hypotheses); $AB^2 + BC^2 + CA^2 = 5R^2$	$5R^2$ case cannot be dropped (affine)
Blundon, Steiner–Lehmus	degenerate triangle	—

9 Does the statement mean what it says?

The verification battery checks that proofs are accepted and gives partial evidence against vacuous statements; it does not show that a Lean statement says what the prose says. To test that, we removed every code name, code listing and link from the mathematical sections of the paper and gave the remaining prose to a language model that had not worked on the developments, with instructions to read nothing else. (Its archived tool log shows that the only file it read was the prepared text; it also compiled a short probe of Mathlib names, and its own statements, on the x86-64 machine; the text was produced before Section 2.4 was written, so Theorem 5 is not covered.) It wrote twelve Lean statements: one for each theorem, two for Theorem 2 (the existential and the explicit form) and two for Theorem 4 ($n = 4, 5$), with every notion it needed defined from the prose, and it marked the places where it found the prose ambiguous. A second model, different from the writer, then tried to prove in Lean, for each pair, that the published statement implies the blind one and conversely. In these proofs it could use the premise, unfolding of definitions, and general Mathlib lemmas, together with the small bridge lemmas about the developments' own notions (for example `Clifford.transversal_iff` and `Lester.bridge_apexOut`), but never the published theorem itself, so that the check is about the statements and not about their truth. We recompiled the comparison files ourselves (no `sorry`, no axioms) and checked that no published theorem is used in a proof.

Ten of the twelve pairs are equivalent in both directions: Theorems 1 and 4 (both parts), the existential part of Theorem 2, and Theorems 6 to 11. The equivalences are not all literal. Lester's theorem was stated over $\mathbb{R}^2$ with apexes defined geometrically and is proved in $\mathbb{C}$ with explicit apexes; the blind Kosnita statement uses lines and "O is not the midpoint of a side" where ours uses collinear triples and non-collinearity; the blind chain predicate is defined by recursion on objects rather than sets. The two remaining pairs are the explicit-point parts of Theorems 2 and 3. Our prose does not give the formula of the explicit point `cp`, so the blind writer quantified over *every* common point of the four circles, as its ambiguity note records. Each direction then holds once one missing fact is supplied as a premise: that `cp` is a common point (from blind to ours),

or that the common point is unique (from ours to blind). We have not proved uniqueness. So for these two theorems the check found a real gap, though in our prose rather than in the Lean: the prose does not determine the Lean statement. Theorems 2 and 3 are about the explicit point cp of Section 2, whose defining formula is in the source and not in the text; read as statements about every common point, they would claim more than we prove. For Theorem 3 the writer also had to guess which three points of each five-point set are designated; its guess agrees with ours for all six sets. Apart from this underspecification, no difference between prose and Lean was found. The blind text, the blind statements, the writer's tool log, the comparison files, our recompilation log and both models' reports are archived.

10 Related work

Formal proofs. The Archive of Formal Proofs entry on Miquel's theorem [16] proves both the incidence form of the three-circle (pivot) theorem (a point on two of the circles lies on the third) and the existence of the common point; Rocq's HighSchoolGeometry [17] proves the incidence form of the four-line Miquel theorem; and a community Lean repository [18] proves the incidence form of the pivot theorem. These three served as a control for our searches. Mathlib's list of 1000 theorems [22] lists Clifford's circle theorems, the five circles theorem, Lester's theorem, Thébault's theorem and the Steiner–Lehmus theorem, all without a formalisation; Kosnita and Blundon are not on the list.

On 13 October 2026 we searched for each of the classical theorems formalised here (Theorems 1–4 and 6–11; the predicate-specific Theorem 5 was not a search target): Mathlib (code, `docs/100.yaml` and `docs/1000.yaml`, issues and pull requests); the full Archive of Formal Proofs [23] (a mirror of 7 October 2026 with 1045 entries, by full-text search, plus the live front page for newer entries); all 1500 articles of the Mizar library; full clones of HOL4 and HOL Light; Metamath's `set.mm`; Rocq's GeoCoq and HighSchoolGeometry; ProofAtlas (sitemap and search index); the public archive of the Lean Zulip; compfiles and `formal-conjectures`; the lean-genius repository [18] (full tree and pull requests); and GitHub code search. In this scope we found no formalisation, complete or partial, of Theorems 1–4 and 6–9, and no complete formal proof of Blundon's inequalities; the nearest is a Lean proof of Gerretsen's weaker inequality $s^2 \le 4R^2 + 4Rr + 3r^2$ [20]. For Steiner–Lehmus the algebraic core is proved in Lean [19] (Section 7), and the Acorn library states the theorem without a proof [21].

The search has limits. The Zulip search covered the public archive only, up to about 2 October 2026; the live search needs a login. GitHub's code index does not cover every repository, returns at most about 50 results per query, and rate limits forced us to use queries not restricted by language, which we filtered by hand. ProofAtlas has no public API, so we used its own index and a third-party `site:` search. The absence of a hit is a statement about this scope only.

Machine proofs outside proof assistants. Wu's method and its successors prove many of geometry theorems by computer algebra, among them Thébault's problem [15]. Java Geometry Expert lists Steiner–Lehmus among its examples, and a Gröbner-basis prover (Hasclid) has an example file for a Blundon-type bound, which we did not run. None of these is checked by a proof kernel. In Wu's method the non-degeneracy conditions are produced by the method [15]; our statements state them explicitly.

Our earlier papers. Papers A [26] and B [27] formalised other classical results by the same coordinate-certificate method, and Paper C [28] treated Barbier's theorem and the four-vertex

theorem.

11 Verification

Gates and tampering tests. The gates pass with only the three standard axioms: `clifford` 20 declarations, `cliffordchain` 32, `lester` 19, `thebault` 12, `kosnita` 11, `fivecircles` 6, `blundon` 5, `steinerlehmus` 3 (108 in all). Each tampering suite plants a hidden `sorry`, an `axiom` split across two lines, a fake `#print axioms` report, a command-redefining macro, a wrapper axiom, and one or more broken hypotheses or bridges that should make the build fail; every planted defect was rejected. The table records the final receipts. On 14 October 2026, on each machine, a script cloned each of the eight repositories from GitHub, checked out the commit pinned in this paper (and recorded that the working tree was clean), recorded the Lean and Mathlib revisions and the SHA-256 hash of every library file, built the library with only the pinned Mathlib supplied, and ran the gate and the tampering suite. The two logs are archived.

development	x86-64 Linux	arm64 macOS
`clifford`	20 / tamper OK	20 / tamper OK
`cliffordchain`	32 / tamper OK	32 / tamper OK
`fivecircles`	6 / tamper OK	6 / tamper OK
`lester`	19 / tamper OK	19 / tamper OK
`thebault`	12 / tamper OK	12 / tamper OK
`kosnita`	11 / tamper OK	11 / tamper OK
`blundon`	5 / tamper OK	5 / tamper OK
`steinerlehmus`	3 / tamper OK	3 / tamper OK

Entries are the number of declarations for which the gate passed, and whether every planted defect was rejected. All sixteen checkouts were at the pinned commit with a clean working tree; the 24 library-file hashes agree between the two logs and with the published commits, and every build used Mathlib `0df444a3` (recorded separately, because a bug in the script left that field of the logs empty). In the first run the script checked out the previous commit of `cliffordchain`, which differs from the pinned one only in a text file; that development was re-run at the pinned commit on both machines with a corrected script, with the results shown, and both runs are archived. (Earlier gate runs, on intermediate versions, are also archived but are superseded by these.)

Independent re-verification. For six of the eight developments (both Clifford developments, `fivecircles`, `lester`, `thebault`, `kosnita`) a separate agent, not the one that wrote the proofs, read the statements against the classical theorems hypothesis by hypothesis and re-checked them numerically; its reports are archived. Their scripts were not kept, so we re-did the numerical part reproducibly for all eight developments, with a single script that computes each configuration from coordinate formulas (for Lester it follows the Lean development's orientation and apex conventions, so it cannot detect a convention error there; the conclusion is symmetric in F_1, F_2 anyway), fixed random seeds, 60-digit arithmetic, and, for every theorem except Steiner–Lehmus, a deliberately false variant as a negative control on the same configurations (script, seeds and output archived with SHA-256 checksums). Results, as configurations / worst residual / false variant violated:

theorem	configs	worst residual	negative control violated
Clifford $n = 4$	1000	1.1×10^{-58}	1000/1000
Clifford $n = 5$	500	1.2×10^{-57}	500/500
Clifford $n = 6$	100	1.1×10^{-57}	100/100
Five circles	500	1.8×10^{-59}	500/500
Lester	1000	1.6×10^{-58}	1000/1000
Sawayama–Thébault	300	2.4×10^{-60}	300/300
Kosnita	1000	1.2×10^{-59}	1000/1000
Steiner–Lehmus	200	7.8×10^{-62}	—
Blundon	1000	(min. margin $4.5 \times 10^{-11} \geq 0$)	310/1000

Each configuration satisfies the hypotheses of the Lean statement up to a margin (pairwise and triple non-degeneracy of the input circles, and of the output triples the script tests, at least 10^{-3} for Clifford; the designated triples of Theorem 3 are non-collinear, being distinct concyclic points, but are not held to that margin; scalene by at least 10^{-2} for Lester, and so on). The Steiner–Lehmus configurations were found by solving $BE = CF$ for the third vertex from random starting points; every solution found was isosceles. For Blundon the residual column gives the smallest relative margin by which s^2 lies inside the two bounds, and the control is the false bound obtained by dropping the square-root term, which fails in 310 of 1000 triangles. (In the agents' earlier runs, one five-circles control held in 1 of 1000 trials; in the re-run every control except Blundon's was violated in every trial. For Clifford $n = 6$ the script checks non-collinearity of three of the five points of each C_S, not of the three designated in the Lean statement; the designated ones follow, since the five points are distinct and concyclic.) Numerical checks are evidence, not proofs.

Statement-level checks. Every development was then run through the statement-level battery used in the earlier papers, on 13 and 14 October 2026: `clifford` on the x86-64 machine and the other seven on the arm64 machine. The battery ran on the sources as they stood on 13 October; since then only comments (docstrings) of `lester` and `kosnita` have changed, with line numbers preserved. `cliffordchain` was re-run on 14 October at the published commit, including the bridge and classification files, and the figures below for it are from that run. The gates and tampering suites were re-run on the final sources on both machines.

- *Conclusion mutations.* The mutation generator alters conclusions of the forms it supports (equalities negated, numeric literals changed) and re-checks the proof. It produced 656 mutants; 618 were rejected by an error, 36 by the heartbeat limit (which shows only that the mutated proof did not complete within the limit), and two survived, both still true: in an auxiliary sign lemma of `thebault`, the conclusion $\sigma X < 0$ weakened to $\sigma X < 1$; and in `cliffordchain`, CliffordChain(1) changed to CliffordChain(2), which also holds by definition (the classification proof that uses the lemma then failed to build). Per development: `clifford` 129/129, `cliffordchain` 283/284, `lester` 82/82, `thebault` 64/65, `fivecircles` 36/36, `kosnita` 33/33, `blundon` 21/21, `steinerlehmus` 6/6. *Most main theorems produced no mutants*, because their conclusions (membership, `Collinear`, `Concyclic`, disjunctions and existentials of these) have no form the generator alters: `Clifford.clifford_one`, `Clifford.clifford_two`, `Clifford.clifford_two_explicit`, `Clifford.clifford_three`, the second development's `clifford_one`, `clifford_two` and `Clifford.clifford_two_clean`, `Lester.lester`, `Thebault3.sawayama_thebault`, `Fivecircles.fivecircles` and `Kosnita.kosnita`. The mutants that were generated test mainly the auxiliary algebra, and `Blundon.blundon` and `SteinerLehmus.steiner_lehmus`. (The `cleanand` tampering

test of the second development does check by hand that replacing the disjunction of `Clifford.clifford_two_clean` by a conjunction breaks the build.)

- *Contradiction probe.* For each declaration with hypotheses, automation tries to derive `False` from the hypotheses alone. In every development the corrected probes found no contradiction. This does not show that the hypotheses are consistent. (A first run of `clifford` reported 24 contradictions. All were proof checks killed by the operating system for lack of memory, which the probe had misread as success; 87 mutation checks were killed in the same way and left unclassified. We changed the harness to count any abnormal exit as an error, removed those 111 records, and re-ran them with less parallelism. The pre-fix result file, which contains the 24 and the 87 records, is archived with the reports.)

- *Satisfiability.* The battery tries to certify, by a Lean witness, that a declaration's hypotheses can hold together. Of 362 declarations, 241 were certified: 57 with hypotheses, 150 whose binders are hypothesis-free, and 34 closed statements (the split is taken from the per-declaration result files); the other 121 are recorded as unresolved. The closed statements include `Clifford.cliffordChain_four`, `Clifford.cliffordChain_five` and `Clifford.Counter6.not_cliffordChain_six`, for which "certified" is trivial; the classification theorems are certified too, which says nothing beyond their being stated. Every main theorem of Sections 2–7 *with* hypotheses is among the unresolved. So the battery does not certify that the hypotheses of any main theorem can be satisfied. The numerical re-check above exhibits configurations satisfying all hypotheses of each main theorem (with the margins stated there); this is evidence, not a certificate.

- *Hypothesis ablation.* Each hypothesis *binder* is removed in turn, and we check whether the unchanged proof still goes through. This tests binders, not their parts: a bundled hypothesis such as `GenPos5` (many conditions in one binder) or the Fermat-point predicates (three line memberships each; the Lester proof uses two of the three) counts as one. In `cliffordchain`, `lester`, `thebault`, `fivecircles`, `blundon` and `steinerlehmus` every tested binder was needed. In `kosnita` two binders of the auxiliary lemma `cert_den` were not (they carry a leading underscore for that reason). In `clifford`, `Clifford.clifford_one`, `Clifford.clifford_two` and `Clifford.clifford_three` needed every binder (10, 20 and 41), but `Clifford.clifford_two_explicit` did not need five of its ten non-collinearity hypotheses (`n135`, `n145`, `n235`, `n245`, `n345`; it has twenty hypotheses in all), the auxiliary lemma `five_NN` six of its, and `ZD_ne_zero` four (all are non-collinearity hypotheses for triples involving circle 5 or 6; we also confirmed the first two cases by reading the proofs). These theorems are stated under more hypotheses than their proofs use; they remain correct. A binder the unchanged proof needs is not thereby shown to be logically necessary.

- *Linter.* The unused-variable linter reports no warnings in the library files of all eight developments as battery-tested (for `cliffordchain`, at the published commit).

These checks show that the proofs are accepted and use only the standard axioms, and give evidence that statements and hypotheses are not vacuous or idle. They do not show that a formal statement means what the prose says. That question is addressed only partly: by the independent re-verification above and by the blind re-formalisation of Section 9 and the author's statement-by-statement check.

Statements checked by hand. The author read each of the fourteen main statements as given in this paper against its Lean statement at the published commit (a sheet pairing the two, with

links to the source lines, is archived with the reports) and found them to agree. This is a reading check, not a proof; Section 9 explains why the explicit-point theorems must be read with the explicit point.

12 Limitations and further work

1. **Clifford's chain for general** n is not proved. The finite-plane predicate we wrote for it is false for every $n \geq 6$ (Theorem 5); a corrected statement, on the Riemann sphere or for n lines, is formulated in prose only. Clifford's theorem itself is proved for $n \leq 6$, and for $n = 6$ only when the five-point circles are not lines.

2. **Euclidean, not inversive.** All statements are about Euclidean circles; lines enter only as stated alternatives. An inversive-plane formulation would remove several exclusions.

3. **Existence.** Lester and Sawayama–Thébault assume the Fermat points and the Thébault circles; we do not prove that they exist.

4. **Blind checks** establish equivalence of statements, not correctness of the prose; the blind writer's isolation is shown by its archived tool log, not enforced; two of the twelve comparisons are conditional on a fact (uniqueness of the Clifford point) that we have not proved; and Theorem 5 was not blind-checked.

Acknowledgements and use of AI

A large-language-model AI assistant was used to write the Lean proofs, to run the prior-art searches and the verification battery, and to draft this paper. The author directed the work and reviewed the paper. Every proof is checked by the Lean kernel, and the author takes responsibility for all content.

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